



A hybrid mathematical model for diabetic data classification using machine learning and deep learning algorithms

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Abstract

The ability of human body to process glucose is affected by diabetes, and this condition can lead to various complications. If left untreated, it can cause various health issues, such as blindness, kidney failure, nerve damage, and amputation of the legs and feet. People with diabetes typically suffer from a reduced quality of life due to their condition and strict diet restrictions. Early diagnosis can help prevent or delay any complications, and it can facilitate treatment adjustments and lifestyle modifications. In addition, it can lower health costs and improve a person's life expectancy. One of the most common methods of diagnosing diabetes is by measuring blood sugar levels. However, this method can be very invasive and uncomfortable. Some other methods only detect the condition after the symptoms have already developed, which can prevent early intervention. Deep learning and machine learning algorithms can help predict the likelihood of a person developing diabetes before its symptoms appear. They can also help improve the accuracy of the diagnosis and provide more comfortable and personalized care for patients. The present study proposed a mathematical model to detect the diabetics by understanding the disease and normal conditions using machine learning and deep learning algorithms. The study proposed a hybrid mathematical model that describes the step-by-step process to perform the classification of diabetic and no-diabetic data classes through feature extraction method. The classification process is conducted through experimentation using Pima Indian Diabetic

dataset. The results achieved has shown that the proposed model has produced an efficient classification with nearly and more than 90% accuracy using machine learning and deep learning techniques respectively. The algorithms have shown an improvement of 20% when the classification is performed with augmented dataset. The future direction of the research is to improve the reliability and accuracy of identifying the diabetic disease using diverge datasets.

Keywords Deep learning algorithms · Diabetic disease classification · Machine learning algorithms · PIMA Indian diabetic dataset

1 Introduction

The chronic disease known as diabetes involves the body's inability to regulate glucose levels, which can lead to high blood pressure and blood sugar levels. If the condition is not treated properly at the early stages, it can cause other complications such as kidney failure and heart disease. A study conducted by researchers indicated that by 2045, the prevalence of diabetes is expected to reach 10.9% of the global adult population. It further estimated that by 2030, the number of people diagnosed with the condition is expected to reach over 600 million (Saeedi et al. 2019). Medical personnel handling diabetes cases require the development of intelligent systems that can help them diagnose and provide decision support.

One of the most common factors that prevent people from being diagnosed with diabetes is the lack of accurate laboratory tests. This is because the process of performing these tests can be time-consuming and costly. Besides this, other factors such as transporting and storing blood samples can also add up to additional expenses. Having a more effective and less invasive diagnosis procedure can help prevent patients from developing complications. Machine learning is a popular choice when it comes to developing systems

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that can perform accurate and timely diagnosis of diabetes. This type of system uses statistical techniques instead of high-speed computation.

Although machine learning systems are capable of distinguishing between linear and nonlinear functions, they are not ideal for identifying risk factors that cause diabetes. This can be solved by adding kernels to the models and using statistical methods to predict such functions. Another type of system that can be used to perform this task is deep learning. This type of system can be developed using a deep learning framework that has high computing power. It can perform well by identifying and solving various functions with minimal manual optimization (Dargan et al. 2020). The advantages of using deep learning systems over machine learning are their low complexity learning functions and robust implementation. A hybrid model that combines the two methods is suggested to allow a system to perform well by analyzing the attributes of the extracted data.

A conclusion is drawn that the present study has combined successfully a hybrid mathematical model with automatic diabetic disease classification using supervised machine learning and TabNet deep learning algorithms. A Pima Indian Diabetic dataset is used and applied preprocessing technique like scaling to adjust the data for proper learning and generalization. The study has applied eight machine learning techniques like logistic regression, support vector machine, random forest, XGBoost classifier, decision trees, naïve bayes, stochastic gradient descent, k-nearest neighbour algorithms, and TabNet deep learning algorithms. Finally, through experimentation approach, it has been observed that the random forest supervised learning algorithm is the best one to attain higher classification accuracies on the selected dataset like original and augmented.

The significant contributions of the study are:

- Proposed a novel mathematical model to implement ML and DL algorithms to perform the classification of diabetic disease data.
- Implemented SMOTE technique to handle the data imbalances.
- Performed hyperparameter tuning with TabNet model structure to adjust parameter values and achieve better results.
- Implemented data augmentation technique to increase the input dataset size and improve the classification performance.
- A comparative analysis is conducted to assess the best suitable ML model to generalize the dataset during classification task.
- Performed the classification task using TabNet architecture which is suitable to understand the tabular data.

The entire paper is organized as several sections like Sect. 2 to discuss the existing literature and summarize the findings, Sect. 3 to propose a hybrid mathematical model to conduct the classification task, Sect. 4 to discuss the experiments conducted, results obtained, interpretations, and open challenges followed by Sect. 5 to discuss the conclusion and future directions.

2 Background

This subsection investigates and profoundly discusses the impact of different machine learning (supervised) and deep learning algorithms in detection and classification of diabetics. This literature review aims to explore the state-of-the-art research on diabetic disease classification using machine learning and deep learning techniques, focusing on articles published in reputable journals during the years 2019 and 2024. Diabetes Mellitus, a chronic condition characterized by the body's inability to effectively regulate blood sugar levels, has become a significant public health concern worldwide. The rapid rise in the prevalence of diabetes is attributed to various factors, including population growth, aging, urbanization, decreased physical activity, and adverse dietary changes. One of the common complications associated with diabetes is diabetic retinopathy, a condition that affects the retina's blood vessels, potentially leading to vision loss.

Early detection and accurate classification of diabetic eye diseases are crucial for timely intervention and effective management. Recent advancements in machine learning and deep learning techniques have shown promising results in the automated detection and classification of diabetic eye diseases using fundus images (Sarki et al. 2020; Maistry et al. xxxx). Several studies have investigated the application of machine learning and deep learning algorithms for the computer-aided diagnosis of diabetic retinopathy. These studies have explored various techniques, such as convolutional neural networks, image processing, and fuzzy logic, to develop automated systems for the detection and grading of diabetic retinopathy and other diabetic diseases (Heydon et al. 2021).

One study has proposed an automated system for the detection of diabetic eye diseases using deep learning on fundus images. The authors developed a robust diagnostic system by integrating state-of-the-art deep learning techniques, including convolutional neural networks, to detect the severity of diabetic retinopathy. Another study reviewed the developments in computer-aided diagnosis of diabetic retinopathy, highlighting the use of machine learning and fuzzy logic techniques for image processing and classification (Pachouly et al. 2022). Furthermore, a comprehensive

review on automated systems for severity grading of diabetic retinopathy and macular edema has been conducted, emphasizing the role of computer-aided diagnosis, convolutional neural networks, and deep learning in these applications (Mathews and Anzar 2021).

Deep Learning and Machine Learning models rely on large datasets to perform well. Unfortunately, it is not always easy to obtain these types of data due to the cost and time involved. While non-invasive techniques can help in some cases, it is not always the best solution. Before creating a dataset, it is important to analyze the link between the target feature, and the diabetic outcome (Wee et al. 2024). Currently, most data-driven models for detecting diabetes are trained using the PIDD, which is a repository of information about the Pima Indians (Dua and Graff 2017). One of the most common methods (Abdulhadi and Al-Mousa 2021; García-Ordás et al. 2021; Kumar et al. 2020) used to eliminate the missing data is by filling in the missing value with the mean value of the various classes. However, this method could not represent the true distribution of the data. This means that when new samples are added to the model, the classification will not be accurate.

Another common method is to remove all the missing data from the model. This will reduce the number of available samples. The boundary between the non-diabetic and diabetic outputs will also be inaccurate if the models have been trained with this dataset (Kumar et al. 2020). In a study conducted on the PIMA Indian dataset, the authors investigated the effectiveness of the feature selection process. They found that the method was able to improve the accuracy of the model to 89.02%. This is a significant improvement over the previous method, which used the correlation score methodology. The authors believe that the improvement was due to the use of the F-Score feature selection strategy, which is related to the PCC process (Lukmanto et al. 2019).

The researchers used the Fisher score method to evaluate the 19 features in the dataset. They were able to identify the features that were most suitable for the classification process. The results of the study indicated that the method can be used to improve the performance of the model and provide clinicians with a quick and accurate diagnosis of diabetes (Dargan et al. 2020). The researchers used the PIDD dataset as the training platform for two deep learning models and two machine learning models. They were able to achieve an accuracy of 79.26% and 73.94%, respectively. The two ML models, namely the RF and SVM, were used in the study (Yahyaoui et al. 2019).

The reviewed articles demonstrate the potential of machine learning and deep learning in improving the accuracy and efficiency of diabetic disease classification. The integration of these advanced techniques with medical data has shown promising results in the early detection and

management of diabetic diseases, ultimately contributing to better patient outcomes.

3 Proposed mathematical model

The main objective of this paper and the subsection is to develop a novel mathematical model to perform the classification of diabetic diseases using machine learning techniques like logistic regression, support vector machine, random forest, XGBoost classifier, decision trees, naïve bayes, stochastic gradient descent, and k-nearest neighbour algorithms and deep learning algorithms like convolutional neural networks, recurrent neural networks, and long-short term memory model. The mathematical modelling involves different unique steps like translating the problem statement into a set of mathematical expressions. The expressions define the data preprocessing activities, feature extraction process, algorithm training, and classification task. The proposed model also assists in evaluating the performance of the algorithms with respect to several metrics like accuracy, precision, recall (sensitivity), and f1-score.

Let us begin the classification task to determine what features determines the diabetic and what features are not. The input data consists of N samples each with d features. Let the input matrix is $X \in \mathbb{R}^{N \times d}$ where X_i represents the feature vector of a patient with $[X_1, X_2, \dots, X_N]$ and $y \in \{0, 1\}^N$ be the corresponding data labels like 0 for non-diabetic and 1 for diabetic. Perform the preprocessing like normalizing the features X as $(X - \mu)/\sigma$ where μ is the X mean vector and σ represents the standard deviation vector. Perform the feature extraction by selecting the significant features either using Pearson correlation coefficient or principal component analysis (PCA) by reducing the data dimensionality using $X_{reduction}$ as $X'W$ where W represents the eigen vector.

The feature transformation of the TabNet architecture is composed of input layer as T_0 with X' and T_0 is the initial representation of the feature. At each decision step, the transformed features are achieved by using fully connected layers (FC) and a non-linear activation function (ReLU). To select the focused features, TabNet uses a sparse attention mechanism at each decision stage M_k as $\text{Sparsemax}(w + b)$ where ' w ' is the weights and ' b ' is the bias for each step. Obtain the selected features T_k for the next decision step as M_k and T_k as an element-wise multiplication. Refine the predictions at each step by accumulating information through each decision step as P_k with a sigmoid function and a sum of ' w ' and ' b '. The final prediction is obtained by combining all the outputs from each decision step with learnable coefficients and each decision step weights. Finally, the binary cross-entropy is used to perform the classification. The 'Adam' optimizer is used to update the weights for a

Table 1 Dataset description

Dataset #	Dataset name	Data features	Data size	Augmentation
Original	ODS	8	768	No
Augmented	ADS1	8	+300	Yes
Augmented	ADS2	8	+2100	Yes

learning rate ‘ η ’. The advantage of TabNet architecture is its attention mechanism that helps to identify important features at every decision step, allows the network to select the relevant features and improve generalization. The networks sparse attention mechanism allows directly on tabular data. Evaluate the performance of each algorithm using the metrics such as accuracy as $(TP+TN)/(TP+TN+FP+FN)$ where TP, FP, TN, and FN are true positives, false positives, true negatives, and false negatives respectively are calculated.

3.1 Mathematical equations

3.1.1 Logistic regression

$$p(y_i = 1|X_i) = \frac{1}{1 + \exp(-(\beta_0 + \beta^T X_i))}$$

where β represents ‘ w ’ and ‘ b ’ model parameters.

3.1.2 Support vector machine

$$w^T X + b = 0$$

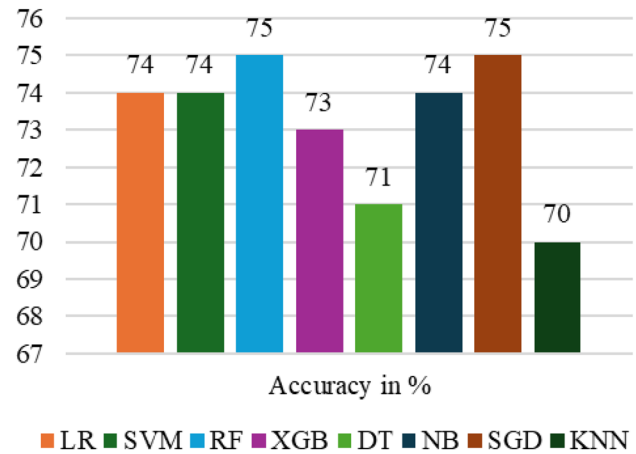
Find the hyperplane to separate the data with ‘ X ’ input data points, ‘ w ’, and ‘ b ’ parameters.

3.1.3 Random forest

Prediction for ‘ X ’ at each tree to determine the final prediction class as $\hat{y} = \text{mode}(\hat{y}^{(1)}, \hat{y}^{(2)}, \dots, \hat{y}^{(M)})$ where M represents the decision tree, and mode denotes the class label that is occurring frequently among the predictions.

Table 2 Classification results with ODS

Algorithm	Accuracy	Precision	Recall	F1-score	Classifications	
					Correct	Incorrect
LR	74	71	70.5	71	170	61
SVM	74	71.5	71	72	172	59
RF	75	73	72.5	72.5	174	57
XGB	73	70.5	71	70.5	168	63
DT	71	69	70.5	69	164	67
NB	74	72	72.5	72	172	59
SGD	75	72.5	72.5	73	173	58
KNN	70	66.5	66.5	66.5	162	69

**Fig. 1** Performance of eight ML models with ODS

3.1.4 XGBoost classifier

Optimize the predictions using gradient boosting as

$$g_i = \frac{\partial l(y_i, \hat{y}_i^{(t-1)})}{\partial \hat{y}_i^{(t-1)}}$$

3.1.5 Decision tree

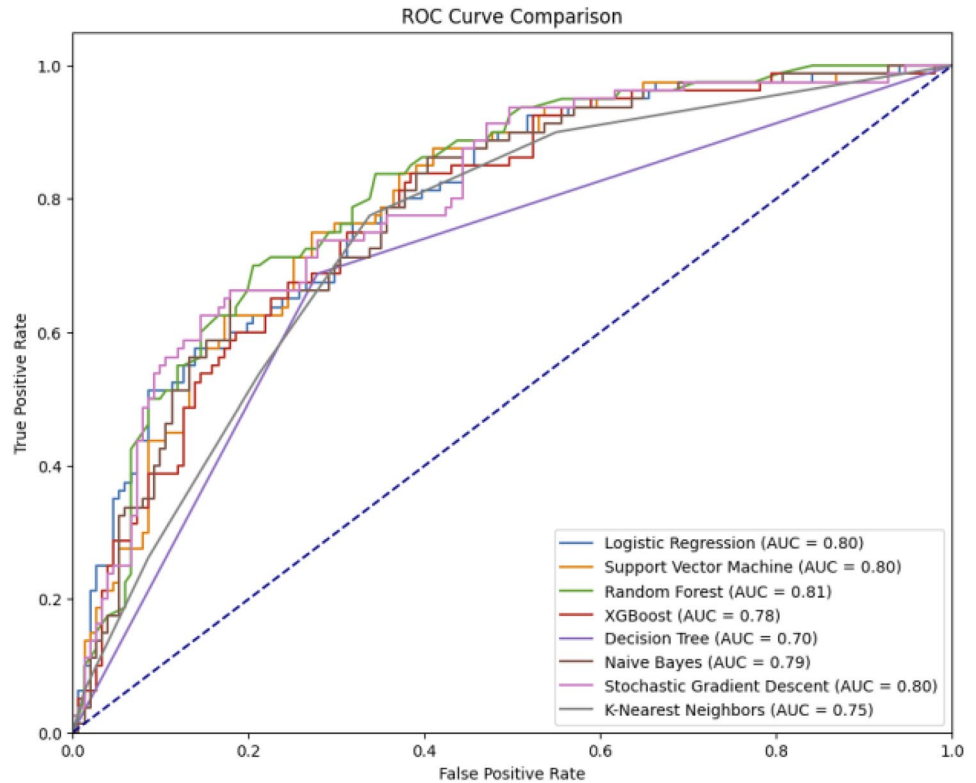
The $\hat{y}$ predicted class for a sample class is defined as $\hat{y} = \text{mode}(y_{leaf})$ to avoid overfitting and perform the classification task.

3.1.6 Stochastic gradient descent

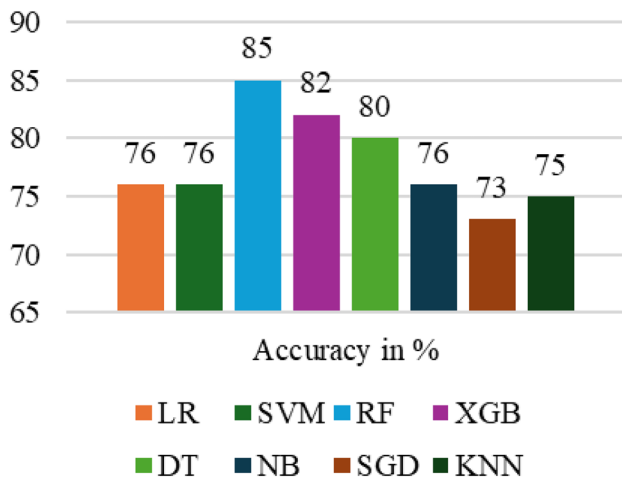
Compute the gradient using (X_i, y_i) for single training example as $\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L(\theta_t; X_i, y_i)$ where $L(\theta_t; X_i, y_i)$ is the loss function for a specific (X_i, y_i) .

3.1.7 Naïve bayes classifier

The final prediction $\hat{C} = \arg \max_c (P(C) \prod_{i=1}^n P(X_i/C))$ maximizes the product of likelihood and the prior probability of the features.

Fig. 2 ROC curve of eight ML algorithms with ODS**Table 3** Classification results with ADS1

Algorithm	Accuracy	Precision	Recall	F1-score	Classifications	
					Correct	Incorrect
LR	76	76	72	73.5	182	58
SVM	76	75.5	74	74	183	57
RF	85	85.5	84	84	205	35
XGB	82	82	81	81.5	198	42
DT	80	79	79.5	79	192	48
NB	76	75.5	72.5	73.5	182	58
SGD	73	72	72.5	72.5	176	64
KNN	75	73.5	72.5	73	179	61

**Fig. 3** Performance of eight ML models with ADS1

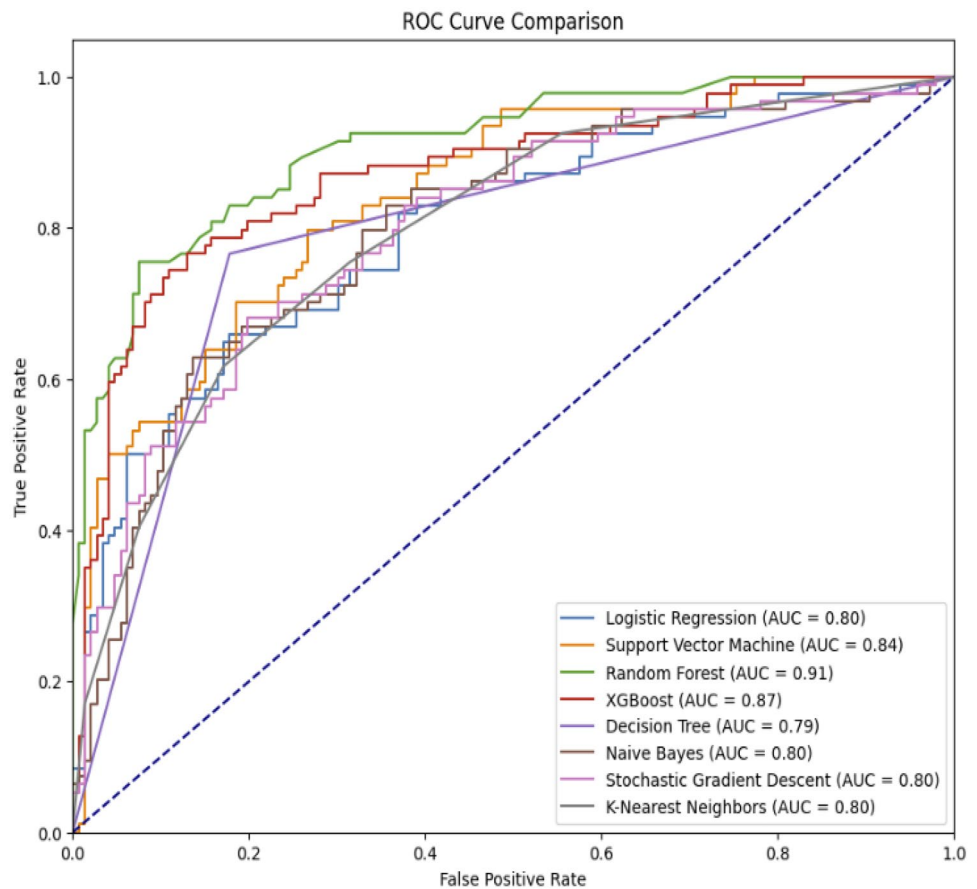
3.1.8 k-nearest neighbour

Classify the new data point by predicting its class as $\hat{y}_q = \text{mode}(\{y_i : X_i \in N_k(X_q)\})$ where $N_k(X_q)$ represents the (X_q) nearest neighbours and mode the neighbours common class label.

4 Experimental approach

4.1 Dataset description

The most reliable and original PIMA Indian Diabetic Dataset (Available Online: Pima Indians Diabetes Database (kaggle.com) xxxx) is used to perform the diabetic (data value 'true') and no-diabetic (data value 'false') data class

Fig. 4 ROC curve of eight ML algorithms with ADS1**Table 4** Classification results with ADS2

Algorithm	Accuracy	Precision	Recall	F1-score	Classifications	
					Correct	Incorrect
LR	86	82.5	68	71.5	669	114
SVM	89	91.5	72	77	691	89
RF	95	97	87.5	91.5	741	39
XGB	95	97	87.5	91.5	741	39
DT	95	97	86.5	91	739	41
NB	82	71	72.5	71.5	636	144
SGD	83	73.5	69	71	649	131
KNN	90	91	76.5	81	702	78

classification using machine learning algorithms and deep learning algorithm. The dataset used for this study was originally from the NIH's Digestive and Kidney Diseases division. The goal of this project is to develop a diagnostic detection and classification test that can predict a patient's likelihood of having diabetes. Some of the constraints that were considered when selecting these instances were the individuals' Pima Indian heritage. The collected data consists of various medical prognostic factors (8 data features) like time_pregnant, glucose_level, diastolic_pb, triceps, insulin, mass_index, predegree, age and a target variable known as the outcome. The former includes the number

of past pregnancies and the patient's BMI, age, and insulin level (Smith et al. 1988).

Initially, the original dataset consists of 768 records and all of them were considered for the classification task. The implementation of various machine learning algorithms is conducted using same dataset to analysis the impact of dataset size on the classification process. This process will guide to understand the impact of dataset size while obtaining the highest classification accuracy on the unseen data. The experiments were conducted on the original dataset and the augmented one for a comparative analysis and get the optimal results. Table 1 describes the datasets used during the experiments.

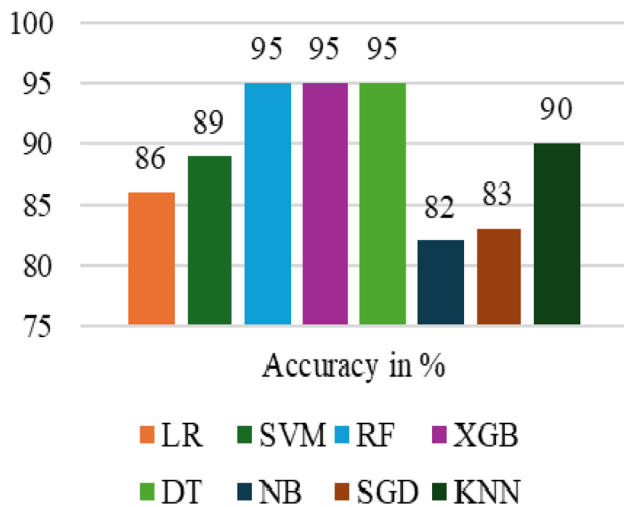


Fig. 5 Performance of eight ML models with ADS2

4.2 Experiments and results

Initially, two experiments were conducted to perform the classification process using original and augmented datasets. A total of eight machine learning algorithms like Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), XGBoost Classifier (XGB), Decision Trees (DT), Stochastic Gradient Descent (SGD), and K-Nearest Neighbour (KNN). Table 2 describes the classification outcomes using the above ML algorithms with original dataset.

Figure 1 shows the performance of all the eight ML models in terms of classification accuracies. It is observed that the RF algorithm is the only one that has conducted the classification well with 174 correct classifications. The KNN algorithm has shown a lowest performance with 69 incorrect classifications along with less precision, recall, and f1score values.

Figure 2 describes the Receiver Operating Characteristics (ROC) curve for eight ML algorithms. Here, the ROC curve is considered to evaluate the performance of the classifier. For a given threshold, the ROC plots the True Positive Rate (TPR) and over the False Positive Rate (FPR). The closer the curve to the top left corner describes the algorithm is better at distinguishing the target class values like diabetic and no diabetic. Area Under the Curve (AOC) in the figure describes the summary measure of the curve's quality.

In general, the AUC value ranges in between 0 and 1, where the 0.5 indicates the random guessing and 1 indicates the perfect classification. It is observed that the RF and DT algorithms have the highest AUC of 81% and 70% respectively whereas the other six algorithms have around 75% to 81%. Table 3 describes the classification outcomes using the above ML algorithms with augmented dataset.

Figure 3 shows the performance of all the eight ML models in terms of classification accuracies. It is observed that

the RF algorithm with the augmented dataset is the only one that has conducted the classification well with 205 correct classifications. The SGD algorithm has shown a lowest performance with 61 incorrect classifications along with less precision, recall, and f1score values.

Figure 4 describes that RF algorithm has achieved a highest AUC at 91% indicating the perfect classification performance followed by XGBoost with 87% and SVM with 84%. DT could be able to manage to obtain just 79% with lower performance compared to other algorithms.

Table 4 describes the classification outcomes using the above ML algorithms with augmented dataset. RF, XGB, and DT algorithms have achieved a highest 95% classification accuracies when compared with other five.

Figure 5 shows the performance of all the eight ML models in terms of classification accuracies. It is observed that the RF, XGB, and DT algorithm with the augmented dataset have performed the classification well with 741, 741, and 739 correct classifications respectively. The NB algorithm has shown a lowest performance with 144 incorrect classifications along with less precision, recall, and f1score values. Figure 6 shows the confusion matrix for RM algorithm with original and two augmented datasets.

Figure 7 describes that RF algorithm has achieved a best AUC at 100% indicating the perfect classification with zero error performance followed by XGBoost with 95%, DT, KNN, and SVM ranging from 86 to 88%. LR and SGD have the lowest AUCs at 80% and less.

4.3 Classification using TabNet

The leveraging deep learning architecture deployed to perform the classification task in the study is the TabNet. The architecture allows to improve the classification performance by learning the tabular data more effectively. The data augmentation technique named SMOTE is used to handling data balancing. It ensures that both the outcome class are presented adequately in the training process and improve the generalization of unseen data during testing. The ROC and the accuracy curves were plotted to visualize how the model has performed during the classification. This process also helps in deciding the need of hyperparameter tuning. The TabNet model is structured with an input layer describing the data features, batch_size and number_of_features. The feature transformer module transforms the data features into meaningful representations. One fully connected layer (dense layer) applied to the input features followed by ReLU activation function helps to introduce the non-linearity in the data. The attention mechanism allows to compute the feature selection and decide which features need focus during each decision. The features that are considered relevant and enabled for further selection process.

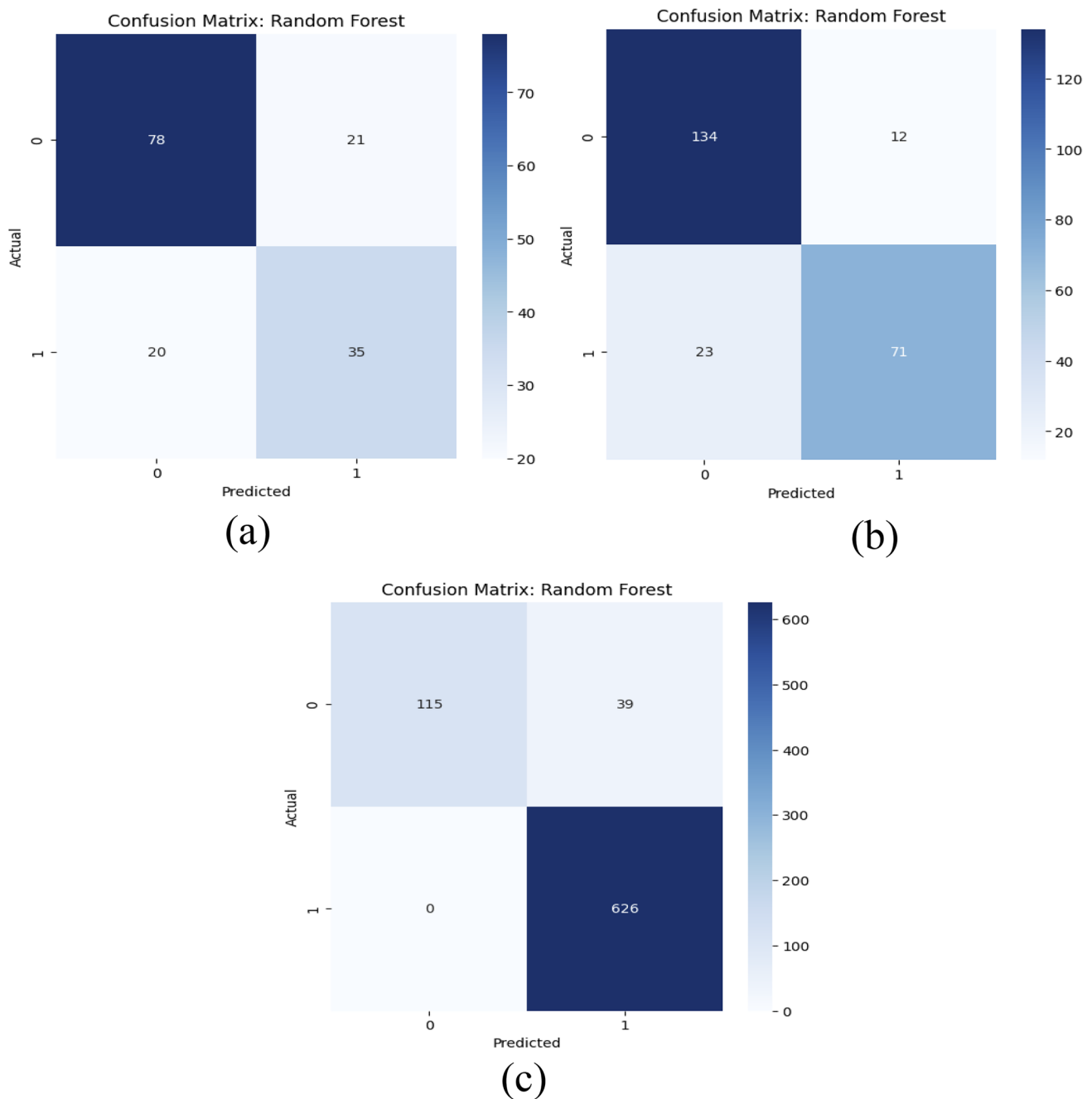
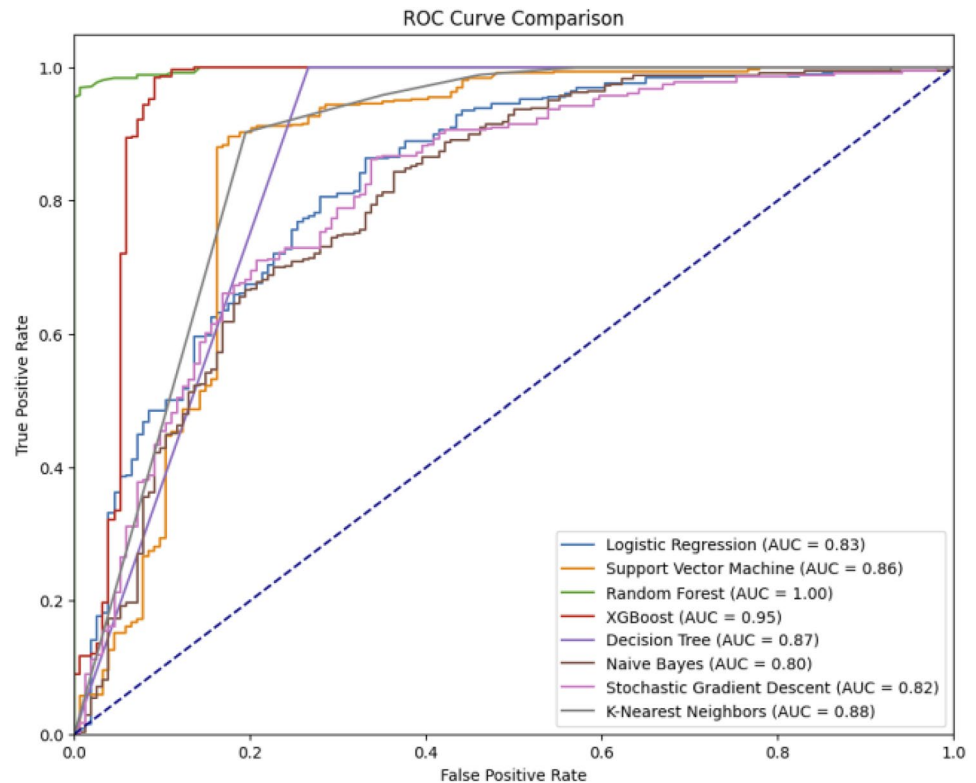
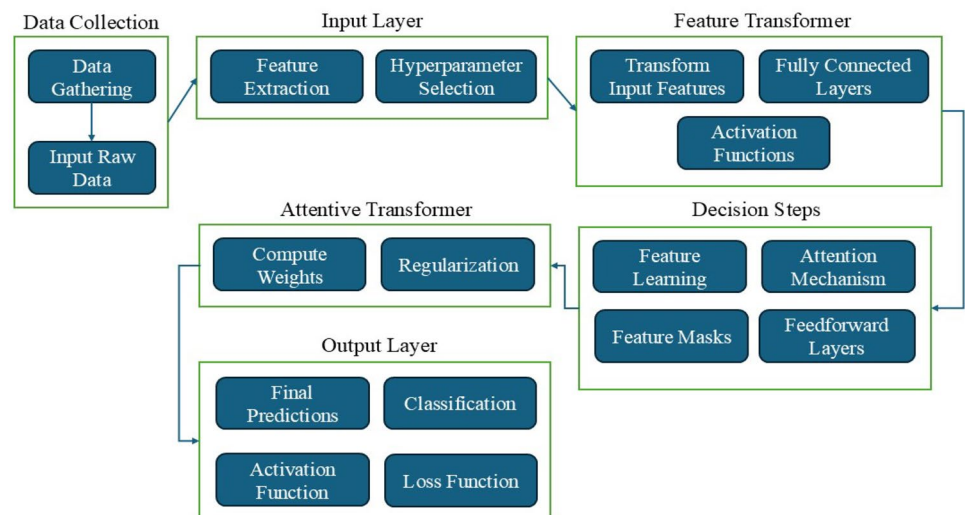


Fig. 6 Confusion Matrix of RM algorithm with datasets **a** original **b** ADS1, and **c** ADS2

The attentive transformer allows to compute the weights of the selected features to improve interpretability and reduce overfitting. The output layer of the architecture produces the final predictions with respect to combined outputs of each attention mechanism. At the output layer, sigmoid activation function is used to classify the 0's and 1's like non-diabetic and diabetic by representing the probability of the positive outcome class. The building blocks of the TabNet architecture is shown in Fig. 8 for better understanding.

The architecture is employed to perform the classification task by splitting the original dataset into training and testing with 80% and 20% data respectively. Initially, at the input layer, 'ReLU' activation function is used to improve the non-linearity in the data. The 'sigmoid' activation function is used at the output layer to perform the binary classification. The experiment is conducted for 100 epochs with a patience value of 10 to handle overfitting of training data. The results obtained after successful training and validation are presented below in Table 5.

Fig. 7 ROC curve of eight ML algorithms with ADS2**Fig. 8** Building Blocks of TabNet Architecture**Table 5** Classification results using DL techniques

Accuracy	Confusion matrix		AUC	Precision	Recall	F1-score
81%	162	38	88%	81	81	81

From Table 5, it is described that an accuracy of 81% indicates that the DL algorithm predicted correctly. An AUC measure of 88% considered to be a good value suggests that the algorithm performs well while differentiating the two target classes. An 81% precision indicates that the algorithm predicts well with a positive outcome and a recall of 81%

indicates the correct identities of the true positive values. An f1-score of 81% indicates a better balance between the precision and recall performing consistently well in detecting the positive cases. Figure 9 shows the ROC curve describing the performance of the DL algorithms with original dataset. The curve nearer to the left top corner indicates the

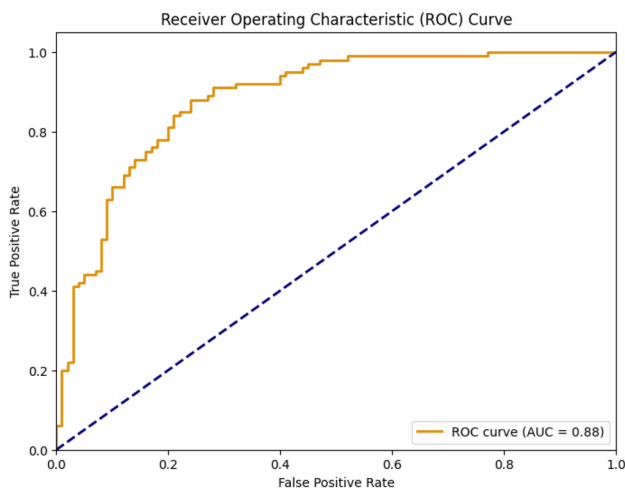


Fig. 9 ROC curve of DL algorithms with ODS

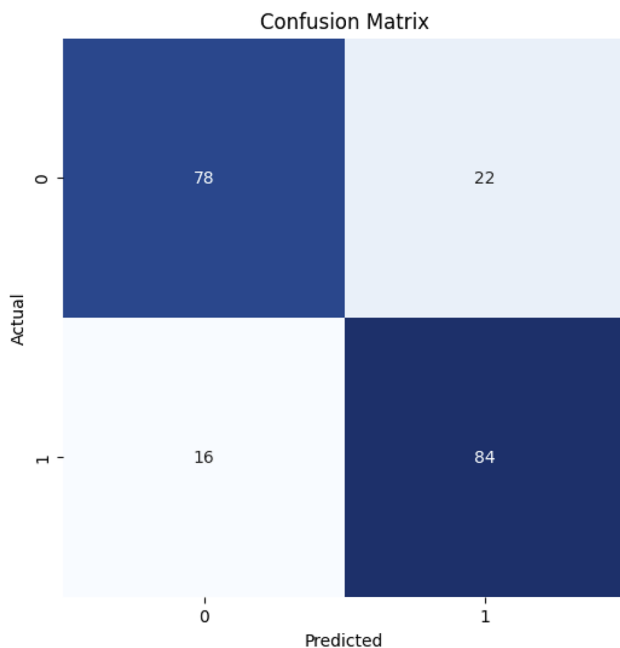


Fig. 10 Confusion matrix for TabNet

effectiveness of the algorithm in differentiating among the diabetic or no-diabetic target class (outcome label) values. Figure 10 shows the confusion matrix for TabNet model during the classification task.

4.4 Interpretations

- Logistical regression and the random forest statistical algorithm performed better than the other approaches with an average AUC of 0.80–0.81 with original dataset.
- The results of the analysis revealed that the decision tree did not perform as well as the other approaches when it came to classification with original dataset.

- The random forest algorithm was the best-performing model when it comes to performing statistical analyses. It had an AUC of 0.91 with augmented dataset.
- The Support Vector Machine and XGBoost exhibited impressive results with respective AUCs of 84% and 87% using augmented dataset.
- The decision tree, on the other hand, had the lowest AUC of 79%. The results suggest that the algorithm might not be as effective at performing this type of task with augmented dataset.
- The random forest algorithm is proved to be the most effective one on the original and augmented datasets with 100% AUC.
- The deep learning algorithm TabNet achieved a good AUC of 88% suggesting it be an effective in differentiating the target classes.

4.5 Open challenges

Despite the progress that has been made in the field of diabetes classification, there are still many issues that need to be resolved. This study aims to provide a comprehensive analysis of the various aspects of this technology.

- A large database is needed to improve the training of models and attain better performance.
- The current diabetes datasets are not representative of the population and may lead to regional or racial biases during training.
- Although certain non-invasive techniques can be utilized in the classification of diabetes, a standard procedure for gathering necessary data is not yet proposed.
- The algorithms for classification will have to be paired with various features and activation functions to produce an acceptable result.

5 Conclusion

Predicting the diabetic disease at the earlier stage can reduce the complications and risks on human health. In this paper, a hybrid mathematical model to perform the classification of binary classes like diabetic and no-diabetic using eight machine learning algorithm, and one deep learning algorithm is implemented. The Pima Indian Diabetic, an open-source dataset has been used to perform the classification by making use of SMOTE technique. This technique is used to handle the data imbalances. The paper reported the best applicable ML algorithm to conduct the classification task using tabular data. The paper examined the performance of each algorithm with respect to accuracy, precision, recall, f1-score, AUC, and confusion matrix. The RF algorithm has

achieved the highest classification accuracy of 75% and 95% with original and augmented datasets respectively. Next, the TabNet deep learning architecture used in this study due to its ability to understand the tabular data using decision tree like interpretability. The unique structure of TabNet allows to learn the more complex relationships among the data while doing right predictions through feature selection process. The architecture has found to be the best tool to handle PIMA Indian Diabetic dataset well by achieving 91% of classification accuracy. In future, the performance of the deep learning model using tabular dataset can be improved by adding more data, tuning the hyperparameters, using ensemble techniques and investigating additional feature engineering. Another extension of the present work is to apply the algorithms on diverge datasets and achieve better results when compared with other state-of-the-art works.

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Declarations

Conflict of interest There is no conflict of interest.

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